# Future work: testing the open boundaries

The staged research program follows directly from the boundaries in sec. 11.13. It separates public-library reproducibility, server theory, portable source-protocol work, task generalization,

# Future work: testing the... (part 2)

and deployment validation so that success in one lane cannot silently certify another.

# Future work: testing the... (part 3)

The order is deliberate. Simulation-study guidance recommends declaring the estimand, data-generating mechanism, and Monte Carlo precision before treating replication as evidence (Morris et al. 2019),

# Future work: testing the... (part 4)

while Monte Carlo error should be reported separately from an interval or a hypothesis test (Koehler et al. 2009).

# Future work: testing the… (part 5)

For nested agents, trials, and seeds, the resampling unit must respect the dependence structure (Loy and Korobova 2021). Accordingly, the phase plan records a primary unit and a falsifier for each extension; a larger sample or more elaborate diagram is not itself a stronger claim.

# Future work: testing the... (part 6)

The implementation registry also separates smoke, pilot, and confirmatory profiles. Pilot worlds select budgets and calibration settings but never enter confirmatory intervals or headline values.

# Future work: testing the... (part 7)

Each completed run must bind its source bundle, configuration, dataset bytes, device, checkpoints, outputs, and completion status into a verifiable receipt. A negative or null scientific result remains a valid citable outcome when those implementation and provenance gates pass; it blocks the intended positive claim,

# Future work: testing the... (part 8)

not the software release.

# Future work: testing the... (part 9)

A separate Friston protocol lane will resolve a machine-readable parity matrix before reconstructing source experiments in Python. Until every required source parameter, routine, unit, and estimand is recovered, that lane will be described as a paper-constrained reconstruction rather than an exact replication.

# Make the sharp server heuristic variational

The asymmetry between the robustness axes (sec. 6.3) motivates a consequential theoretical design question that is explicitly parked outside the authorized roadmap.

# Make the sharp server... (part 2)

The client-side $\beta$/rcce update carries a derived, loss-specific bounded-influence result under the matching assumptions; the sharp server-side `robust_aggregate` carries only its recovery limit (eq. 10).

# Make the sharp server… (part 3)

The new variational aggregator (sec. 8.8.2, sec. 15) supplies a *rigorous* server-side rule — exact block updates descending the stated free energy eq. 12 through non-increasing block updates,

# Make the sharp server... (part 4)

with a proven raw effective-weight bound and the same recovery corner. What it costs is conservatism: it is the maximum-entropy consensus, not the sharp accuracy-maximizer.

# Make the sharp server... (part 5)

That parked question is sharper than before: write down a generalized variational objective in the FedGVI family (Mildner et al. 2025), informed by recent closed-form GVI characterizations (Nguyen and Westerhout 2026), logarithmic-pool weighting theory (Carvalho et al. 2023),

# Make the sharp server... (part 6)

and robust divergence-weighted federated aggregation (Li et al. 2022), whose closed-form minimizer is competitive with the empirical reweighting across declared contamination regimes.

# Make the sharp server… (part 7)

That would combine axis 3's effective-weight bound with axis 2's empirical sharpness in one server rule. The recovery corner and the variational objective together supply two boundary conditions any such unification must satisfy.

# Make the sharp server... (part 8)

Any empirical choice of `robustness` or `entropy_weight` will be made on separate calibration worlds using a proper log score, then frozen before confirmatory evaluation. This guards against selecting an apparent leader with evaluation truth and preserves null or reversed confirmatory outcomes.

# Make the sharp server... (part 9)

The comparison family will include logarithmic and linear pools, the current heuristic, the variational family, and a centered-log-ratio geometric-median control; none inherits a parameter-space robust-federated-learning guarantee merely by operating on beliefs.

# Promote the baseline to original FedGVI scale

The bounded-influence result is anchored locally by the small NumPy logistic-regression baseline (fig. 21).

# Promote the baseline to... (part 2)

Promoting it to the GPU-scale Bayesian-neural-network experiments of the source paper (Mildner et al. 2025) — the experiments deferred here (sec. 11.13) —

# Promote the baseline to... (part 3)

would test whether the per-client robustness curve holds at the model capacity and contamination regimes where federated learning (McMahan et al. 2017) actually operates.

# Promote the baseline to... (part 4)

It would connect the discrete-POMDP result to the partitioned-VI line (Ashman et al. 2022; Bui et al. 2018). The planned comparison would also require posterior-parameterization parity with Bayesian neural-network work, rather than treating the current deterministic point-mass MLP as a posterior (Mildner et al. 2025).

# Promote the baseline to... (part 5)

The portable lane preserves the source protocol's site factors, client cavity, factor-replacement update in natural coordinates. A synthetic CPU/MPS pilot already exercises the cavity-conditioned local optimizer, explicit device and fallback receipts, and checkpoint/resume equivalence.

# Promote the baseline to... (part 6)

It does not establish source-dataset parity.

# Promote the baseline to... (part 7)

The next local campaign distinguishes a locked portable CPU/MPS profile from an exact source-scale CUDA profile that remains external until suitable hardware is available. FashionMNIST anchors source-dataset protocol parity, while MNIST and KMNIST test portability.

# Promote the baseline to... (part 8)

A separate source-bound tabular pack will report proper-score effects per licensed dataset, with training-only preprocessing and byte-, split-, and license-level provenance; nested seeds will not be treated as independent datasets.

# Extend hierarchical federation beyond the current stack

The governing caveat here is that added depth must be shown to *earn* generalization, not merely to execute:

# Extend hierarchical... (part 2)

on the current sentinel task the deeper stacks match rather than beat the flat baseline on location (sec. 11.15). The extension therefore has two distinct fronts —

# Extend hierarchical... (part 3)

carrying the recovery contract to deeper stacks, and finding a task family in which depth actually pays.

# Extend hierarchical... (part 4)

The 2-level hierarchical POMDP (sec. 17.9) couples location inference to a single global context. The generic N-level architecture uses `build_nlevel_world` from the `fedference.pomdp` module and has already been exercised with a 3-level stack (sec. 17.11,

# Extend hierarchical... (part 5)

sec. 17.12): a meta-context variable (L3) gates the context prior (L2) which in turn gates the location prior (L1).

# Extend hierarchical... (part 6)

The empirical-prior top-down messages (eq. 38, eq. 39) remain valid variational steps at every depth, and the log-linear-pool federation at each level is bit-identical to the in-process result (Proposition 12).

# Extend hierarchical... (part 7)

The natural next question is whether the limit-as-proof contract of sec. 11.13 survives still deeper hierarchies: does the recovery corner (context prior $\to$ uniform) remain checkable to machine precision when the L2 $\to$ L1 message is itself a function of an L3 belief coupled to an L4 belief,

# Extend hierarchical... (part 8)

and can message-passing engines such as RxInfer (Bagaev et al. 2023) carry the generic alternating-minimization at scale?

# Extend hierarchical... (part 9)

The current hierarchical diagnostic asks a narrower question. Under one configured surprise threshold, it classifies the non-gating top level below the threshold and the informative top level above it (sec. 18.1). This is an implementation control,

# Extend hierarchical... (part 10)

not a posterior refit or an autonomous depth-selection result.

# Extend hierarchical... (part 11)

A future breadth-and- depth search must define its candidate family, scoring rule, uncertainty, replication units, and out-of-sample evaluation before it can support a model- selection claim for the generic N-level stack.

# Extend hierarchical... (part 12)

The next task family is deliberately controlled rather than merely deeper: partially observable Four Rooms and Key-Door will compare flat, oracle, learned, shuffled, and non-gating hierarchies at matched horizons and compute. Task is the higher-level replication unit. A gain in only one task will remain task-specific,

# Extend hierarchical... (part 13)

and the larger campaign will not begin until the hybrid representation recovery gates pass.

# Move from process transport to true multi-machine federation

The current deployment uses queue-backed single-machine processes and loopback sockets. Promotion to cross-host workers would address the remaining deployment caveat of sec. 11.13.

# Move from process... (part 2)

That extension must preserve the bit-identical consensus property proved in Proposition 12.

# Move from process... (part 3)

The `federation/` package and federation tests already establish the API contract: a server collects serialized beliefs from 5 worker channels, fuses them with `robust_aggregate` at robustness $c = 1.5$, and broadcasts the consensus back over response channels, with bit-identity verified at True.

# Move from process... (part 4)

The loopback socket path adds optional pre-shared-key frame integrity and file-backed digest-verified replay validation. The next systems step is an explicitly local Docker multi-node emulator with mTLS by default, HMAC compatibility, checkpoint/restart, and reproducible message-fault controls.

# Move from process... (part 5)

It is not physical multi-host evidence.

That later claim requires receipts from distinct hosts, deployment-grade key management, timeout policy, and long-running orchestration across process restarts, but it does not require changing the mathematics.

# Move from process... (part 6)

Secure aggregation, differential privacy, and Byzantine tolerance are separate future threat models; transport integrity alone would not establish any of them (Blanchard et al. 2017; Pillutla et al. 2022).

# Move beyond categorical state spaces

This work is discrete-categorical, matching the community's worked POMDP (Da Costa et al. 2020). A first step is already in place: the closed-form Gaussian KL and Rényi divergences of sec. 16 show the divergence family — and its $\alpha \to 1$ recovery — carries over verbatim to Gaussian beliefs,

# Move beyond categorical... (part 2)

scoped out of the categorical experiments.

# Move beyond categorical... (part 3)

Continuous-state Gaussian generative models would test whether the limit-as-proof contract survives the move off categorical state spaces — whether the recovery corner remains checkable to machine precision when the belief simplex is replaced by a Gaussian belief,

# Move beyond categorical... (part 4)

and whether message-passing engines such as RxInfer (Bagaev et al. 2023) can carry the robust update at scale.

# Move beyond categorical... (part 5)

Extending the structure-learning study (Smith et al. 2022; Friston and Penny 2011) into continuous models would, in parallel, test whether robust belief fusion and robust *structure* fusion compose.

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# Move beyond categorical... (part 6)

A minimal executable fixture now gates a discrete dynamics context over continuous position and velocity, Gaussian observations, and bounded actions. The bounded pilot now includes matched naive, robust, discrete-only, continuous-only, and oracle-context components, a singular-covariance rejection,

# Move beyond categorical... (part 7)

and next-position predictive scoring.

# Move beyond categorical... (part 8)

It is still a representation, recovery, and control surface rather than confirmatory task evidence. The full study must freeze calibration and budgets, use independently held-out worlds, retain outlier falsifiers, and execute the preregistered comparison before supporting a hybrid-task claim.